KAVIYATHAMIZHAN T K

Chennai, Tamil Nadu • 9943991074 • kaviyathamizhan37@gmail.com
github.com/Kaviyathamizhan | linkedin.com/in/kaviyathamizhan-t-k

AI/ML Engineer | Agentic AI · LLMs · Edge AI

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Reinforcement Learning, Edge AI, and intelligent system development. Built end-to-end machine learning solutions spanning model optimization, LLM fine-tuning, forecasting, and AI deployment using PyTorch, FastAPI, and modern MLOps practices.

EDUCATION

B.Tech Computer Science and Engineering With AI (IBM-affiliated)

Bharath Institute of Higher Education and Research | CGPA: 9.16/10 *Sep 2022 - June 2026*

WORK EXPERIENCE

Sure Trust | Machine Learning Engineering Intern *Sep 2025 - Mar 2026 | Remote*

- Improved brand monitoring accuracy by fine-tuning BERT for multi-class sentiment classification and emotion detection on social media datasets, delivering structured JSON output via RESTful APIs.
- Built cardiovascular disease risk prediction model with feature engineering and class balancing; evaluated with F1-score and AUC-ROC for reliable diagnostic performance.
- Implemented experiment tracking and model versioning, reducing model comparison time from hours to minutes.
- Built an AI-driven energy optimization platform using LightGBM forecasting (RMSE 0.50, F1 0.786), engineering 16+ temporal features and implementing a SciPy HiGHS-based optimization engine for cost-efficient energy scheduling.

Ramco Cements Ltd. | Data Analyst Intern *Sep 2025 - Nov 2025 | Chennai, TN*

- Accelerated logistics decision-making by preprocessing enterprise datasets and building Tableau dashboards surfacing supply-chain insights to cross-functional teams.

SKILLS

AI/ML & Deep Learning: PyTorch, Transformers, Hugging Face, LoRA, LLM Fine-Tuning, Reinforcement Learning (GRPO), Reward Engineering, Model Compression, Structured Pruning, Knowledge Distillation, RAG, NLP

Backend & Tools: Python, FastAPI, RESTful APIs, SQL, SQLAlchemy, Docker, Git/GitHub, ONNX Runtime, DuckDB

ML & Data Engineering: Scikit-learn, Pandas, NumPy, LightGBM, Feature Engineering, Time-Series Forecasting, Linear Optimization (SciPy HiGHS), Classification, EDA

Frontend & Cloud: React.js, JavaScript, HTML5/CSS3, Tailwind CSS, Microsoft Azure, IBM Watsonx, Weights & Biases

PROJECTS

CrisisGrid - Multi-Agent Disaster Response Intelligence System | CrisisGrid_System

Python, PyTorch, RL, LoRA, Qwen 1.5B, Hugging Face, Weights & Biases (W&B) | Agentic AI · LLM Fine-Tuning · Autonomous Agents

- Developed autonomous agent system for disaster management with tool-use, planning, and memory components using GRPO-based fine-tuning on Qwen 1.5B with LoRA adaptation.
- Implemented reward engineering and agent evaluation framework; investigated reward-alignment challenges in long-horizon multi-agent decision-making under uncertainty.

Pipeline Debug Env - Agentic AI Environment for Data Pipeline Debugging | pipeline_debug_env

Python, FastAPI, DuckDB, Docker, OpenEnv, Pandas, Hugging Face Spaces | Agentic AI · RL Environment · MLOps

- Built an OpenEnv-compatible agentic environment with fault injection, reward grading, and action validation for training autonomous AI agents on multi-step pipeline failure diagnosis.
- Designed reward-grading and action-validation logic enabling agents to recover from multi-step pipeline failures with reduced debugging iterations.

AEDSP - Adaptive Energy-Driven Structured Pruning Framework *Sep 2025 - Apr 2026 | Remote*

PyTorch, ONNX Runtime, CIFAR-10, ResNet-18, MobileNetV2, VGG-16 | Edge AI · Model Optimization

- Achieved 59% model compression and 50%+ FLOPs reduction on deep neural networks by implementing adaptive structured pruning with minimal accuracy loss.
- Developed multi-objective scoring engine (energy cost, inference latency, sensitivity curves) with knowledge-distillation-based recovery and ONNX deployment benchmarking.

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization | DeepLearning.AI - Coursera
- Top 800 Finalist | Meta PyTorch OpenEnv Hackathon x Scaler School of Technology (31,000+ participating teams)